

BRC WRF Michael Practical Test Packet

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This is the short handout for John/Michael pair-programming on the current WRF workflow. It is meant for a no-run walkthrough first, then for planning the first approved practical tests.

It is not approval to run DTN staging, WPS, `real.exe`, `wrf.exe`, Slurm jobs, large downloads, or scaling sweeps. Those steps need a named human approval, scope, and stop point before they run.

Goal For The First Session

By the end of the first session, Michael should be able to explain:

- which repo owns each layer of the workflow;
- where staged input, active runs, and durable archives live;
- which Jan-2013 Basin facts are proven;
- which commands are safe no-run checks;
- which settings are safe to edit in a case manifest; and
- where the first real WPS/WRF approval gate starts.

Current Truth

TOPIC	CURRENT STATE
Proven case	Jan-2013 Uinta Basin, NAM-only, nested 12/4 km, 2013-01-31 12Z through 2013-02-02 00Z.
Proven model path	<code>brc-tools</code> staged inputs, WPS, <code>real.exe</code> , <code>wrf.exe</code> , archive, and no-run quicklooks.
Input identity	NAM-only, <code>Vtable.NAM</code> , <code>interval_seconds = 21600</code> . The old proof scratch used WPS default intermediate prefix <code>FILE</code> .
Not proven	GEFSv12 reforecast plus NAM two-stream forcing with <code>fg_name = 'GEFS', 'NAM'</code> and <code>interval_seconds = 10800</code> .
Current run profile	<code>owned_notch392_max</code> : <code>lawson-np</code> , <code>notch392</code> , one node, 56 tasks, 900G, <code>srun --mpi=pmi2</code> .
Fresh staging contract	Fresh <code>brc-tools</code> staging should emit <code>manifest_<case>.json</code> and <code>contract_<case>.json</code> . This repo currently carries a reconstructed legacy NAM-only contract for strict validation.

Repo Split

REPO	OWNS	DO NOT ADD THERE
<code>brc-tools</code>	NWP download/staging, manifests, contracts, input quicklooks.	WPS, <code>real.exe</code> , <code>wrf.exe</code> , or Slurm run profiles.
<code>brc-wrf</code>	WRF source, WPS/WRF consumption docs, case manifests, validators, Slurm rendering, WRF-output quicklooks.	NWP downloader or GRIB staging logic.
<code>brc-knowledge</code>	Canonical CHPC node, storage, scheduler, proxy, and validated example-script facts.	Repo-local code or case manifests.

Read Before Pairing

Read these in order:

1. `brc-docs/BRC-WRF-STATE-PLAYBOOK.md`
2. `brc-docs/BRC-WRF-FIRST-CASE.md`
3. `brc-cases/README.md`
4. `../brc-tools/docs/WRF-STAGING-STATE-PLAYBOOK.md`

Keep `doc/BRC_WRF_MICROTASK_HANDOFF.md` open as the detailed task board.

No-Run Walkthrough

Run these from `~/gits/brc-wrf`. They read metadata, verify existing files, or render text/plots from existing proof artifacts. They do not submit work.

```
git status --short --branch --untracked-files=all

python brc-cases/wrf_case.py validate \
  brc-cases/jan2013_basin_nam.case.yaml

python ../brc-tools/scripts/stage_wrf_inputs.py --verify-manifest \
  /scratch/general/vast/$USER/wrf_inputs/jan2013_basin_gefs/manifest_jan2013_basin_gefs.json

python brc-cases/wrf_case.py validate \
  brc-cases/jan2013_basin_nam.case.yaml --strict-files

python brc-cases/wrf_quicklook.py check \
  brc-cases/jan2013_basin_nam.case.yaml

python brc-cases/wrf_case.py render-slurm \
  brc-cases/jan2013_basin_nam.case.yaml
```

Optional no-run visual output:

```
python brc-cases/wrf_quicklook.py render \
  brc-cases/jan2013_basin_nam.case.yaml
```

The optional render writes ignored PNGs under `brc-cases/quicklooks/jan2013_basin_gefs/`.

Settings To Inspect Or Change

Use `brc-cases/jan2013_basin_nam.case.yaml` as the first settings map.

SETTING GROUP	KEYS OR FILES	SAFE NO-RUN ACTION
NWP source	<code>forcing.sources</code> , <code>forcing.wps_fg_name</code> , <code>forcing.interval_seconds</code> , <code>forcing.contract_path</code>	Inspect or draft a copied case manifest. Do not claim GEFS+NAM works until WPS evidence exists.
Input destination	<code>paths.input_root</code> , <code>paths.grib_data</code>	Confirm the scratch input path. New staging belongs in <code>brc-tools</code> and may need DTN approval.
Run destination	<code>paths.run_root</code> , <code>paths.wps_run</code> , <code>paths.wrf_run</code>	Confirm the active scratch layout. Do not run WPS/WRF from the git checkout.

SETTING GROUP	KEYS OR FILES	SAFE NO-RUN ACTION
Archive destination	<code>paths.archive_root</code> , <code>archive.required_patterns</code> , <code>archive.colon_safe_wrfout_source</code>	Confirm durable output target and colon-safe <code>wrfout</code> handling.
WPS details	<code>wps.vtable</code> , <code>wps.ungrib_prefix</code> , <code>wps.namelist_fg_name</code> , <code>paths.geog_data_path</code>	Check consistency. WPS execution is a separate approval gate.
WRF runtime	<code>slurm.profile</code> , <code>slurm.ntasks</code> , <code>slurm.memory</code> , <code>slurm.nodelist</code> , <code>slurm.mpi_launcher</code>	Render and review Slurm text only. No <code>sbatch</code> without explicit approval.

First Practical Tests

ORDER	TEST	WHY FIRST	STOP POINT
1	Fresh NAM-only contract validation	Proves a new <code>brc-tools</code> sidecar can replace the reconstructed legacy contract.	<code>wrf_case.py validate --strict-files</code> passes against fresh <code>contract_<case>.json</code> .
2	GEFS+NAM WPS-only field proof	Checks whether the two-stream forcing design has the needed fields.	Stop after <code>metgrid</code> ; show <code>met_em</code> field list, <code>num_metgrid_levels</code> , and warnings. Do not run <code>real.exe</code> .
3	WRF scaling/memory benchmark	Finds whether 16, 28, or 56 tasks and a smaller memory request are enough.	Requires approved <code>sbatch</code> ; record timing, memory evidence, WRF marker, and archive path.

Approval Gates

Before any approved run, write down:

ITEM	VALUE
Approved by	TBD
Approval date/time	TBD
Repo and command scope	TBD
Data source	<code>NAM-only</code> or <code>GEFS+NAM</code>
Input destination	<code>/scratch/general/vast/\$USER/wrf_inputs/<case>/</code>
Run destination	<code>/scratch/general/vast/\$USER/wrf_runs/<case>/</code>

ITEM	VALUE
Archive destination	/uufs/chpc.utah.edu/common/home/lawson-group6/<namespace>/wrf_archive/<case>/run_<UTC>/
Stop point	fresh contract validation , metgrid only , real.exe only , or full WRF
Evidence to preserve	Command, log path, manifest/contract path, WPS/ WRF markers, archive path

Result Table

Use this table during the no-run walkthrough.

CHECK	COMMAND OR FILE	EXPECTED RESULT	OBSERVED RESULT	FOLLOW-UP
Git state	<code>git status --short --branch --untracked-files=all</code>	Clean or intentionally dirty	TBD	TBD
Case metadata	<code>wrf_case.py validate</code>	Pass	TBD	TBD
Input manifest	<code>stage_wrf_inputs.py --verify-manifest</code>	Existing proof manifest verifies	TBD	TBD
Strict case files	<code>wrf_case.py validate --strict-files</code>	Pass against current reconstructed contract	TBD	TBD
Quicklook check	<code>wrf_quicklook.py check</code>	Finds required existing proof artifacts	TBD	TBD
Slurm render	<code>wrf_case.py render-slurm</code>	Text only, no submission	TBD	TBD

Use this table before changing settings.

SETTING	CURRENT VALUE	PROPOSED VALUE	FILE TO EDIT	PROOF NEEDED BEFORE USE
NWP source	<code>nam_analysis</code>	TBD	<code>*.case.yaml</code> and <code>brc-tools</code> staging config	Fresh manifest/ contract and WPS field evidence
WPS <code>fg_name</code>	NAM intent, old proof scratch used <code>FILE</code>	TBD	<code>*.case.yaml</code> , <code>namelist.wps</code> draft	Paired ungrib prefix and metgrid output

SETTING	CURRENT VALUE	PROPOSED VALUE	FILE TO EDIT	PROOF NEEDED BEFORE USE
interval_seconds	21600	TBD	*.case.yaml , WPS/WRF namelists	Cadence matches staged inputs
Tasks	56	TBD	Slurm profile/ template	Approved benchmark
Memory	900G	TBD	Slurm profile/ template	Approved benchmark and peak memory evidence

What To Hand Around

Use these Markdown sources and rendered PDFs as the handout set:

- `brc-docs/BRC-WRF-MICHAEL-PRACTICAL-PACKET.md`
- `brc-docs/BRC-WRF-STATE-PLAYBOOK.md`
- `brc-docs/BRC-WRF-FIRST-CASE.md`
- `brc-cases/README.md`

Recommended PDF output directory:

```
brc-docs/handouts/
```

Render with the local Markdown-to-PDF wrapper:

```
md2pdf brc-docs/BRC-WRF-MICHAEL-PRACTICAL-PACKET.md \
  -o brc-docs/handouts/BRC-WRF-MICHAEL-PRACTICAL-PACKET.pdf
md2pdf brc-docs/BRC-WRF-STATE-PLAYBOOK.md \
  -o brc-docs/handouts/BRC-WRF-STATE-PLAYBOOK.pdf
md2pdf brc-docs/BRC-WRF-FIRST-CASE.md \
  -o brc-docs/handouts/BRC-WRF-FIRST-CASE.pdf
md2pdf brc-cases/README.md \
  -o brc-docs/handouts/BRC-WRF-CASE-SCAFFOLDS.pdf
```